

2025

COMPUTER SCIENCE — HONOURS — PRACTICAL

Paper : DSCC-7P

(Operating System Lab)

Full Marks : 25

Set - 1

Marks Distribution

Source Code	: 10
Output	: 05
LNB	: 05
Viva voce	: 05
	25

Answer *any one* question.

1. Write a shell script to print Pascal triangle.
2. Write a shell script to identify and display the **longest word** present in a given text file.
3. Write a shell script to count the words, lines and characters of a given file. File name should be provided at run time.

(2)

D(4th Sm.)-Computer Science-H/Pr./DSCC-7P/CCF(Set-1)

4. Write a shell script that displays a **real-time digital clock** (HH:MM:SS format) that updates every second for a specified duration.
 5. Write a shell script to find a number from a given array using binary search technique.
 6. Write a shell script that accepts a combination of 5 file names and directory names as command-line arguments. The script should check each argument to determine whether it is a file or a directory and report accordingly. If an argument is a file, display the number of lines it contains.
 7. Write a shell script to perform bubble sort, in ascending order, on an array of numbers. The script should receive a file name as a command-line argument, where the file contains the numbers. The sorted output should be displayed on the terminal.
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COMPUTER SCIENCE — HONOURS — PRACTICAL

Paper : DSCC-7P

(Operating System Lab)

Full Marks : 25

Set - 2

Marks Distribution

Source Code	: 10
Output	: 05
LNB	: 05
Viva voce	: <u>05</u>
	25

Answer *any one* question.

1. Write a shell program to print Fibonacci series within a given range.
2. Write a shell program to reverse a string and check whether it is palindrome or not.
3. Write a shell script that takes a word from user and find out the frequency of the word in a given file.

(2)

D(4th Sm.)-Computer Science-H/Pr./DSCC-7P/CCF(Set-2)

4. Write a shell script to print all the Armstrong numbers from 1 to 500.
 5. Write a shell script to count the number of vowels in a given string.
 6. Write a shell script which prints the contents of a text file from line number m to line number n , where m and n are to be supplied as command line arguments.
 7. Write a shell script to reverse a string.
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2025

COMPUTER SCIENCE — HONOURS — PRACTICAL

Paper : DSCC-7P

(Operating System Lab)

Full Marks : 25

Set - 3

Marks Distribution

Source Code	: 10
Output	: 05
LNB	: 05
Viva voce	: 05
	<u>25</u>

Answer *any one* question.

1. Write a shell script to sort a list of numbers using Selection sort.
2. Write a shell script to remove duplicates from a given array of integers.
3. Write a shell script to convert a given integer (in days) to years, months and days, assuming that all months have 30 days and all years have 365 days.

(2)

D(4th Sm.)-Computer Science-H/Pr./DSCC-7P/CCF(Set-3)

4. Write a menu driven program to store book details like title, author and price. Write a shell script program to input details for five books. Find the most expensive and the lowest priced books and display their information.
 5. Write a shell script for matrix addition and multiplication.
 6. Write a shell script to print prime numbers up to a given range.
 7. Write a shell script to check permission of a file.
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2025

COMPUTER SCIENCE — HONOURS — PRACTICAL

Paper : DSCC-7P

(Operating System Lab)

Full Marks : 25

Set - 4

Marks Distribution

Source Code	: 10
Output	: 05
LNB	: 05
Viva voce	: 05
	<u>25</u>

Answer *any one* question.

1. Write a shell script that searches for all occurrences of a specific string (provided as the argument) in a text file named 'document.txt'.
2. Write a shell script to display all words of a file in ascending order.

(2)

D(4th Sm.)-Computer Science-H/Pr./DSCC-7P/CCF(Set-4)

3. Write a program in shell script to display a pattern like a diamond as below.

```
  *
 ***
*****
*****
*****
*****
*****
  *
  *
```

4. Write a shell script to find a number from a list of integers.
5. Write a shell script to count the no. of words, lines and characters of a given file.
6. Write a shell script to sort a set of integers using bubble sort technique.
7. Write a shell script to find GCD of two numbers.
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